

Manuel MADEIRA

CONTACT

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ABOUT ME

Final-year PhD candidate at EPFL, advised by [Pascal Frossard](#) and [Dorina Thanou](#). My research focuses on discrete generative models, particularly how to improve controllability and align training with inference in diffusion and flow-matching methods. Using graph generation and language modeling as testbeds, I study both how generative models leverage the inherent structure of data and how structural and domain knowledge can be explicitly incorporated into generative processes, with applications including molecular generation and digital pathology.

EDUCATION

- 2022 - PRESENT PhD in MACHINE LEARNING (Expected Graduation: December 2026)
 École Polytechnique Fédérale de Lausanne, Switzerland
Research on structurally controlled graph generation, discrete diffusion and flow matching, and training–inference alignment, with applications including molecular generation and digital pathology.
- 2018 - 2021 MSc. in BIOMEDICAL ENGINEERING
 Instituto Superior Técnico, Portugal
GPA: 19 / 20 (**1st** in class)
THESIS: *Using Co-Coercivity for Variance Reduction in Stochastic Convex Optimization* (**20/20**)
SUPERVISORS: [Pedro Aguiar](#), [João Xavier](#), and [Renato Negrinho](#)
- 2018 - 2019 Exchange student in COMPUTER SCIENCE
 (FALL) **Tsinghua University**, China
- 2015 - 2018 BSc. in BIOMEDICAL ENGINEERING
 Instituto Superior Técnico, Portugal
GPA: 19 / 20 (**1st** in class; first student in the program to graduate with this GPA)

EXPERIENCE

- APR 2026 - PRESENT **Research Intern** at APPLE MLR
Investigating training-inference alignment for discrete diffusion models.
- JUL 2025 - DEC 2025 **Applied Scientist Intern** at AMAZON
Developed a graph-based deep learning framework achieving an approximately 1000× efficiency improvement over existing tools for supply chain long-term planning.
- SEP 2021 - AUG 2022 **Machine Learning Researcher** at [INDUCTIVA RESEARCH LABS](#) (Startup)
Conducted research on deep-learning-based approaches to solve partial differential equations. Deployed physics-informed neural networks to model heat diffusion and coastal dynamics and analyzed their generalization to arbitrary domains.
- MAR 2021 - JUN 2021 **Research Internship** at [INSTITUTE FOR SYSTEMS AND ROBOTICS](#)
Extended my MSc thesis work on exploiting L -smoothness in first-order stochastic optimization, contributing theoretical developments to algorithmic analysis.
- SEP 2019 - FEB 2020 **Research Internship** at [IST-ID](#)
Transcribed interviews on health technology assessment in Portugal for the [MEDI-VALUE PROJECT](#).

AWARDS AND HONORS

- 2025 **NeurIPS 2025 Top Reviewer**
- 2025 **Qualcomm Innovation Fellowship Finalist**
- 2022-23 **EDIC PhD Fellowship**, EPFL
- 2016-18 **1st-ranked student across all years of the Biomedical Engineering BSc**, Instituto Superior Técnico
- 2015-20 **Diploma of Academic Excellence (Top 10%)**, Instituto Superior Técnico
- 2009-15 **Best Student in High School Award (monetary prize)**, Crédito Agrícola

PUBLICATIONS

* Equal contribution

- [1] S. Rozada, Y. Qin, MM, P. Frossard, and A. Ribeiro. **DiPhon: Diffusion on Graphons for Scalable Graph Generation**. Under review, 2026.
- [2] S. Ögüt, C. Vincent-Cuaz, MM, et al. **Benchmarking Instance-Level Learnability and Interpretability in Multiple Instance Learning**. Under review, 2026.
- [3] A. Carballo-Castro, MM, Y. Qin, D. Thanou, and P. Frossard. **Generating Directed Graphs with Dual Attention and Asymmetric Encoding**. In: *NPGML Workshop*. NeurIPS. 2025.
- [4] B. Chen*, C. Vincent-Cuaz*, L. A. Schoenpflug*, MM*, et al. **Revisiting Automatic Data Curation for Vision Foundation Models in Digital Pathology**. In: Early Acceptance (Top 9%). MICCAI. 2025.
- [5] O. Zaghen, MM, L. Toni, and P. Frossard. **Graph Discrete Diffusion: a Spectral Study**. In: *DeLTa and XAI4Science Workshops*. ICLR. 2025.
- [6] L. Sbicego, S. Ögüt, MM, Y. Qin, D. Thanou, and P. Frossard. **On the Role of Graph Structure in Graph Neural Networks**. In: *ICBINB Workshop*. Selected for PMLR Proceedings. ICLR. 2025.
- [7] Y. Qin*, MM*, D. Thanou, and P. Frossard. **DeFoG: Discrete Flow Matching for Graph Generation**. In: Oral Presentation (Top 1%). ICML. 2025.
- [8] MM, C. Vignac, D. Thanou, and P. Frossard. **Generative Modelling of Structurally Constrained Graphs**. In: NeurIPS. 2024.
- [9] MM, D. Thanou, and P. Frossard. **Tertiary Lymphoid Structures Generation through Graph-based Diffusion**. In: *GRAIL Workshop*. MICCAI. 2023.
- [10] S. Moalla*, MM*, L. Riccio*, and J. Lee*. **[Re] Reproducibility Study of Behavior Transformers**. In: *ML Reproducibility Challenge 2022*. Outstanding Paper Award (Honorable Mention). ReScience C. 2023.
- [11] MM, R. Negrinho, J. Xavier, and P. M. Aguiar. **COCO Denoiser: Using Co-Coercivity for Variance Reduction in Stochastic Convex Optimization**. In: *OPT Workshop*. NeurIPS. 2021.

TALKS

ICML 2025, Vancouver, Canada. *DeFoG: Discrete Flow Matching for Graph Generation*.

AI Alliance: AI in Materials & Chemistry Webinar Series, [online](#). *DeFoG: Discrete Flow Matching for Graph Generation*.

Inductiva Machine Learning for Science & Engineering Summer School, Porto, Portugal, 2025. *An Introduction to Graph Neural Networks*.

BioML Workshop, Lausanne, Switzerland, 2025. *Constrained Graph Generative Models in Science*.

Applied Machine Learning Days, Lausanne, Switzerland, 2025. *Constrained Graph Generative Models in Science*.

DataMakers Fest, Porto, Portugal, 2023. *Generative Models in Science: A Peek into Graph Diffusion Models for Digital Pathology*.

TEACHING EXPERIENCE

TEACHING: Network Machine Learning (three editions; Lead TA in 2024/25 and 2025/26)
Probability and Statistics
Practice of Object-Oriented Programming
Analysis I
Histology

SUPERVISION: Six students, including one junior PhD student

SKILLS

PROGRAMMING: Python, Java, MATLAB, Mathematica, R
ML & DATA: PyTorch, PyTorch Geometric, Lightning, TensorFlow/Keras, NumPy, scikit-learn, Pandas
OPT. & RESEARCH: Hydra, Weights & Biases, Git, CVXPY, CVXOPT
INFRASTRUCTURE: AWS (EC2, S3), Docker, Kubernetes, Slurm, Run:AI
LANGUAGES: Portuguese (native), English (C2; TOEFL iBT: 114 / 120), Spanish (A2), French (A2), Chinese (elementary; Tsinghua course)

COMMUNITY SERVICE

REVIEWING: JOURNAL: IEEE Transactions on Signal and Information Processing over Networks (2026)
CONFERENCE: NeurIPS (2025)
WORKSHOPS: NPGML (NeurIPS 2025); ComBayNS (IJCNN 2025)
OUTREACH: Through [Native Scientists](#), introduced AI, generative modeling, and diffusion models for Portuguese-speaking students from migrant backgrounds in Switzerland (ages 6–18)

EXTRACURRICULAR ACTIVITIES

2023	Attended Cambridge Ellis Unit Summer School on Probabilistic Machine Learning
2022 - 2024	Co-organizer of Deep Learning Sessions Portugal
2021	Attended Lisbon Machine Learning Summer School
2009-21	Competitive football player (national level)
2015	National finalist in Biology Olympiad

PERSONAL INTERESTS

Football, CrossFit, Skiing, Reading, Cinema